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`Mike Konczal`

4 Alternate Model

We have a new model, of the form:

1. $y_t = \beta_0 + \beta_1 z_t + \beta_2 p_t + \beta_3 (p_t)^2 + \beta_4 p_t z_t + u_t$

The following results are:

	coef	std error	t-stat
β_0	-7.39147	0.20321	-36.373
z_t	2.57118	0.07764	33.115
p_t	-2.57368	0.18581	-13.851
p_t^2	-0.27127	0.03754	-7.227
$p_t z_t$	0.71702	0.05931	12.090

4.1 Intuition

What to make of $\beta_2 < 0, \beta_3 < 0$, and $\beta_4 > 0$.?

4.2 Optimal P

With a bit of algebra we can get: $\eta = -1 => p_t = \frac{-1(\beta_2 + \beta_4 x_0 + 1)}{2\beta_3}$

4.3 Squared Term

(c) What to do about this square term? Here is a plot of the squared term using the FWL Theorem to isolate the squared term:

This is not visibly compelling for keeping this term.

Let's use what we've learned about information criterion to see if this term is helpful. We check the BIC criterion for model with the square term against the model without it (we re-estimate it and regress), and find that there is more information for keeping it. So we do.

4.4 Revenue Maximizing

$x_0 = 15 => ln(max_p) = .6783594 => max_p = 1.97$

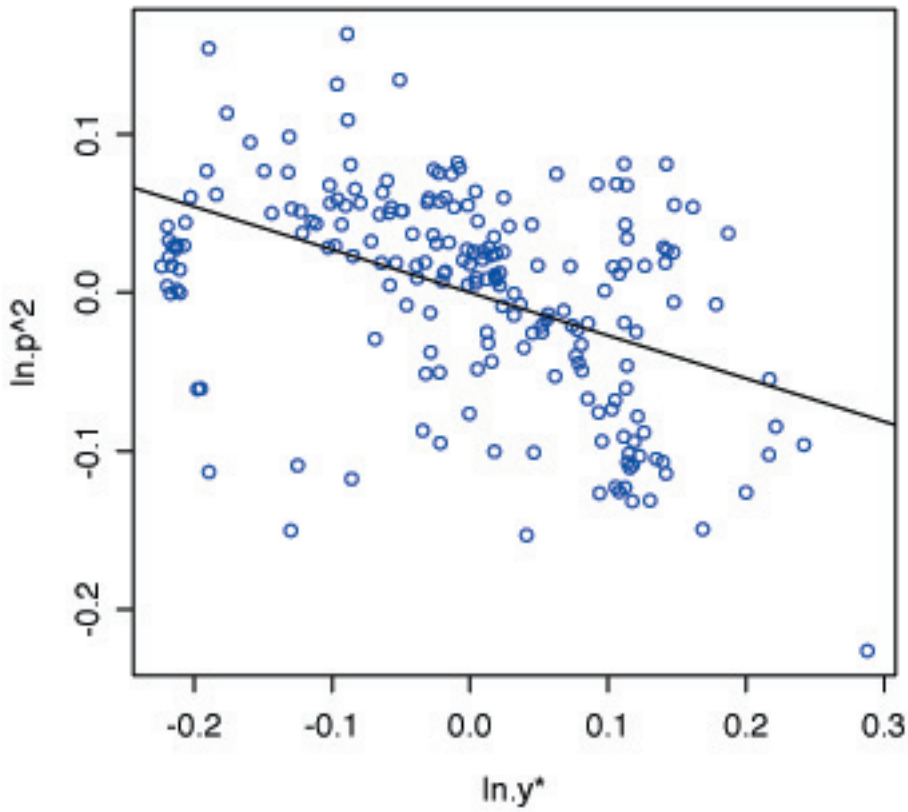
4.5 Confidence Interval

We then estimate a confidence interval using the δ method, the bootstrap off of the residuals, and the bootstrap off of the coefficients (each bootstrap run 10,000 times):

	std dev	lower bound	upper bound
δ	0.1601	-0.1537	0.474
$residuals$	0.168	0.434	1.0992
$coefficients$	0.2039	0.4006	1.196

4.6 Critical Questions

1. Why the low price? 95th percentile of price = 1.10
2. Oligopoly failed?
3. Monopsony
4. A question for history and political science?


ln.p^2
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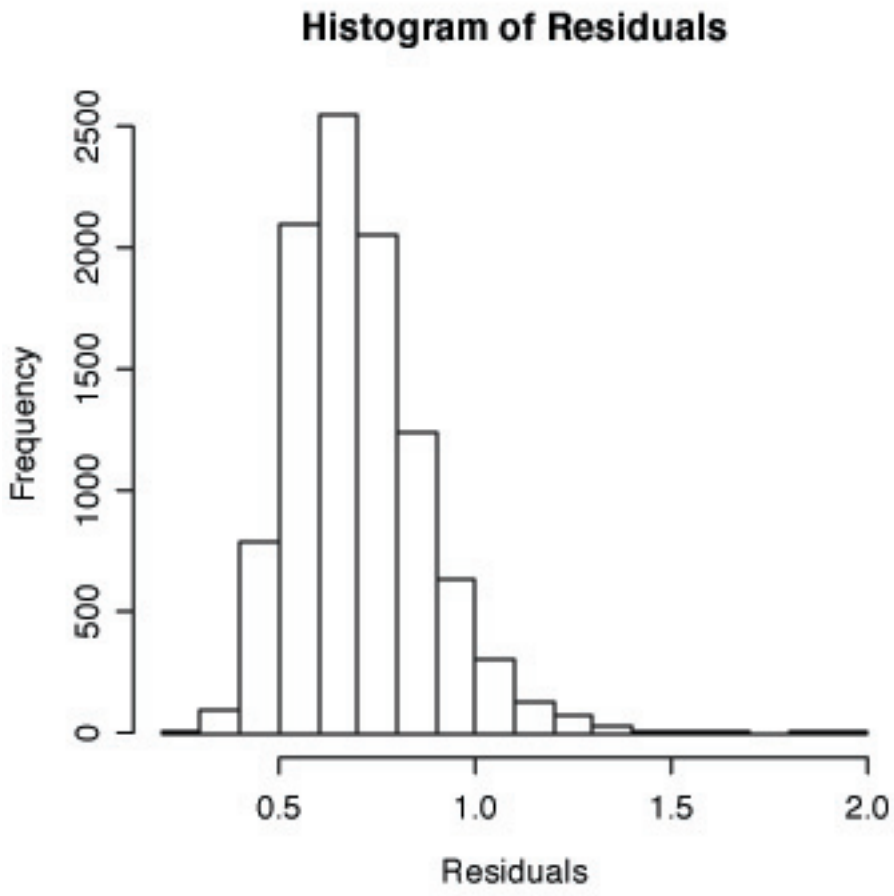
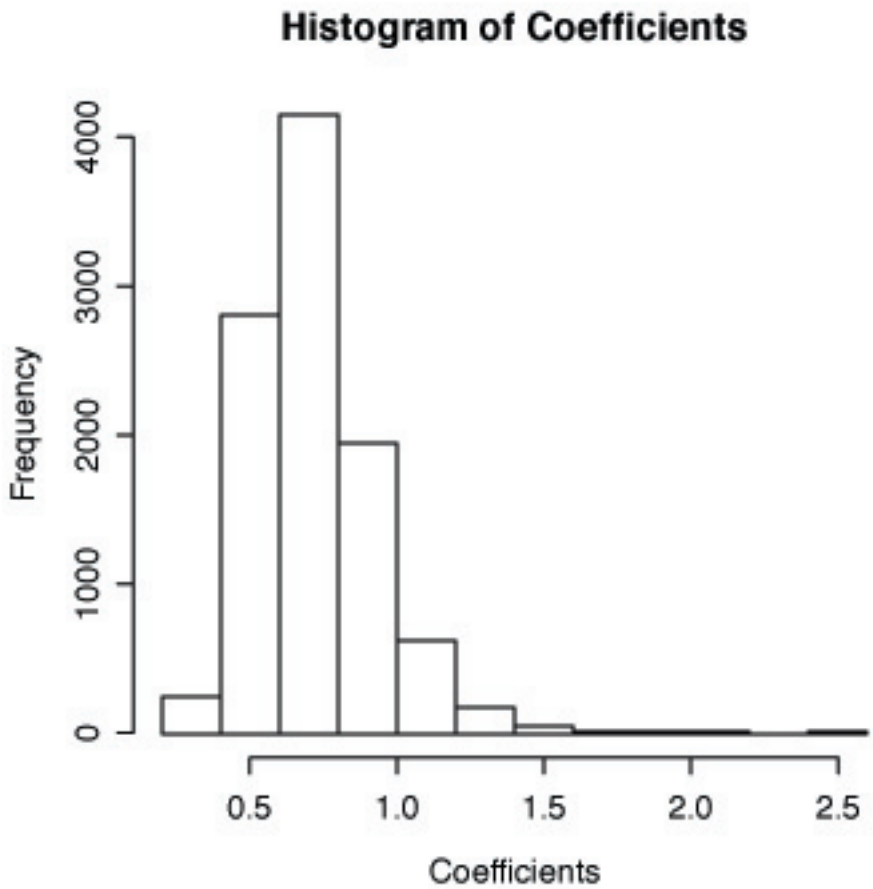
0.0
-0.1
-0.2
-0.2
-0.1
0.0
0.1
0.2
0.3
ln.y*


Histogram of Residuals
Frequency
0
500
1000
1500
2000
2500
0.5
1.0
1.5
2.0
Residuals


Histogram of Coefficients
Frequency
0
1000
2000
3000
4000
0.5
1.0
1.5
2.0
2.5
Coefficients